

SOP-STATIC-5100: Static Electricity Bonding and Grounding

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Section 1. Purpose and Scope

Governs bonding and grounding controls to prevent static discharge ignition during transfer of flammable liquids and handling of combustible dust.

Section 2. Bonding Requirement

Clause 2.1 — Bonding between a delivery vehicle (tank truck/rail car) and the receiving system is required before any flammable liquid transfer connection is made, and shall remain connected until the transfer is complete and lines are disconnected.

Clause 2.2 — Bonding cable/clamp continuity shall be verified (visually or with a continuity tester where required by the specific transfer procedure) before transfer begins — a bonding clamp attached to a painted or corroded surface may not achieve actual electrical continuity.

Section 3. Container Filling

Clause 3.1 — Filling a portable container with a Class I flammable liquid from a bulk source requires bonding between the container and the fill nozzle/source; filling an ungrounded plastic container with a flammable liquid from a metal drum or tank is prohibited regardless of bonding to the source, due to the container's own insulating properties.

Section 4. Fill Rate

Clause 4.1 — Initial fill rate into an empty vessel/container is limited to reduce splash-charging (typically below 3 ft/s until the fill pipe is submerged), per the specific transfer procedure's rate limit.

Section 5. Dust Handling

Clause 5.1 — Equipment handling combustible dust (conveyors, dust collection systems) shall be bonded and grounded per its area classification; a dust collection system serving a combustible dust process requires continuity verification at commissioning and annually thereafter.

Section 6. Records

Clause 6.1 — Bonding/grounding continuity test records are filed to the Process Safety File and retained for 3 years.